Appendix A. Methodology

Papers were collected from Google Scholar, Semantic Scholar, and arXiv using combined keywords such as "text-to-image", "t2i" "generated images", "fairness", "bias", "representation bias", "stereotyping", "demographic bias." Backward and forward snowballing were used to expand coverage. Inclusion criteria: papers written between 2022-2025 directly related to text-to-image that measured, analyzed, or mitigated fairness/bias. Exclusion criteria: general fairness papers in computer vision without clear text-to-image relevance. Limiting factors were English-only papers. For each paper, we extracted model type, dataset, prompts, demographic attributes, evaluation metrics, findings, and mitigation methods. Fairness type was determined by the paper's main evaluation target, (demographic, proportional, performance parity). This was then divided into target/threshold fairness as defined earlier. Bias type was determined by the source or manifestation of the problem and from there sorted into categories.

Appendix B. Extended Evaluation Strategies

This appendix provides the full evaluation-strategy detail summarized in Section 4, organized by evaluation space.

B.1. Manual audits

Manual audits include human counts, qualitative reviews, and user studies of T2I outputs.

User studies In user studies (interviews, focus groups, open-ended generation tasks), participants recognize representation and stereotyping biases in demographic (gender, race, age, disability) and contextual (occupation, countenance, socio-economic, cultural) attributes, which they describe as disparaging or otherwise socially detrimental (Mack et al., 2024; Qadri et al., 2023; Ghosh et al., 2025; Apiola et al., 2024; Ghosh et al., 2024; AlDahoul et al., 2025).

Manual audits (which involve author counts/reviews) often deal with occupation and education contexts which are compared to real-world workforce/enrollment statistics (proportional representation) (Szymański et al., 2025; Gengler, 2024; Currie et al., 2024) Other contextual attributes arise here: facial expressions, multi-subject composition, agency, activities, and censorship (Sun et al., 2024; York et al., 2024), which in turn yields composition, fidelity, and censorship bias (Sandoval-Martin and Martínez-Sanzo, 2024; Baxter, 2023) In this section, we note the use of target fairness (i.e. real-world statistic matching) (Sandoval-Martin and Martínez-Sanzo, 2024).

B.2. Automated audits

Automated audits includes generated images labeled at scale with pretrained attribute prediction tools (like CLIP), embedding analyses (prompts/images encoded into a shared embedding space) and VQA/caption analyses (vision language tools which label attributes and check caption-image alignment.)

Automated audits Generated labeling typically uses demographic parity (equal representation between binary gender and six skin tones) (AlDahoul et al., 2025) and proportional representation through occupation statistics (Cheong et al., 2024). Intersectional analyses here are usually limited to gender and race (Seshadri et al., 2024; Perera and Patel, 2023).

Embedding analyses Focuses mainly on gender and race/skin tone, expanding into various objects and contexts (D'Inca et al., 2024; Luccioni et al., 2023; Zhang et al., 2023; Teo et al., 2024). As with automated audits, fairness is declared as demographic parity, importantly in our estimation, this is without a well-justified target.

VQA/caption audits Covers gender, race/skin tone, and occupation (Cho et al., 2023; Zhang et al., 2023; Hu et al., 2023; Zhang et al., 2024). We also start to see additional demographic attributes covered, such as disability and personality traits (Bianchi et al., 2023; Naik and Nushi, 2023). Fairness is declared as demographic parity (Zhang et al., 2023), proportional representation Bianchi et al. (2023) or performance parity Zhang et al. (2024). The main bias types are representation and stereotyping.

Across the evaluation space of automated audits, reported biases include representational gaps, amplification, occupational stereotyping, and compositional bias (Lyu et al., 2025; Seshadri et al., 2024; Bianchi et al., 2023; Cho et al., 2023). We saw fairness applied to an expanded range of contextual and demographic attributes, and found several examples of threshold fairness in effect (Naik and Nushi, 2023; Friedrich et al., 2025).

B.3. Benchmarks

Benchmark suites include safety and open-set bias coverage, investigating harmful content and privacy evaluations (Li et al., 2025b). Attributes surveyed include occupation/demographic attributes and broader prompt sets span scenes, objects, clothing, and products (Lee et al., 2023; D'Inca et al., 2024). These works focus mainly on gender, race, age, and include contextual factors like professional–dress pairings and multi-attribute scenes (Bakr et al., 2023). Findings commonly show stereotyping representation skews and amplification, plus trade-offs between fidelity and fairness. Fairness is operationalized as demographic parity gaps in representation/toxicity, proportional representation (Luccioni et al., 2023) and, for open-set methods, performance parity in errors. In this evaluation space, we note the use of target fairness, with Jin et al. (2024) employing threshold fairness.

Appendix C. Extended Mitigation Strategies

This appendix provides the full mitigation-strategy detail summarized in Section 5, organized by intervention point.

C.1. Prompt Engineering (Inference-Time Intervention)

Prompt engineering, or "prompt injection," operates as a lightweight inference-time strategy. By explicitly appending demographic descriptors (e.g., "female doctor" or "Latino engineer") to neutral prompts, these methods override the model's biased default associations (Bonna et al., 2024; Bansal, 2024). Techniques range from manual user injection (Cheng et al., 2024) to automated appending of randomized attributes when neutral professions are detected (Friedrich et al., 2025; Kim et al., 2023).

While effective for black-box models where parameter access is restricted (Hao et al., 2023; Luo et al., 2024), this approach acts as a superficial patch rather than a cure. It shifts the burden of fairness to the user and fails to address the root representational bias within the model's knowledge. Consequently, biased reasoning may persist in downstream tasks or when the model is used without strict prompt guardrails.

C.2. Pre-training and Fine-tuning (Model Realignment)

Biases in T2I models are largely result of *amplification* regarding pre-existing skews in training datasets (Friedrich et al., 2025; Perera and Patel, 2023; Seshadri et al., 2024). Addressing this at the root requires data curation or model retraining, though end-to-end retraining is often computationally prohibitive and curation alone may reduce model generalization capabilities (Nichol et al., 2021).

Recent work favors **Parameter-Efficient Fine-Tuning (PEFT)** to align models with fairness targets without full retraining. Shen et al. (2023) utilize Low Rank Adaptation (Hu et al., 2022) with a *Distribution Alignment Loss* to steer the model toward diverse demographic distributions. Similarly, Concept Sliders by Gandikota et al. (2024a) and Unified Concept Editing (UCE) by Gandikota et al. (2024b) employ low-rank adaptors or closed-form weight updates to surgically transform or edit biased concepts in the cross-attention layers, effectively "unlearning" specific stereotypes while preserving general capability.

C.3. Conditional Embedding Manipulation

Bias often originates in the text encoder, where neutral profession terms act as geometric proxies for specific genders or races (Bolukbasi et al., 2016; Gonen and Goldberg, 2019). Mitigation here focuses on decoupling these associations before they reach the generative backbone.

Geometric Projection: Methods like DebiasVL by Chuang et al. (2023) and Fair-Imagen by Fu et al. (2026) identify bias directions in the embedding space and project profession embeddings onto an orthogonal subspace, effectively "nulling out" the gender signal.

Additive Steering: Alternatively, methods like Fair Mapping (Li et al., 2025a), SANER (Hirota et al., 2025), ITI-Gen (Zhang et al., 2023) and (Li et al., 2024) introduce lightweight trainable modules or learned vectors that add corrective signals to the embedding. Embedding Arithmetic Thirugnana Sambandham and Schön (2026) computes the general demographic attribute direction vectors using vector arithmetic, and then use them to steer the demographic attribute of the generated images while maintaining concept coherence.

Token Optimization strategies intervene at the token level of the text-encoder, optimizing "de-stereotyping" prefixes (Kim et al., 2023) or appended direction vectors (Kim et al., 2025) to encourage diversity without altering the prompt's semantic core.

C.4. Diffusion Process Manipulation

Diffusion manipulation intervenes dynamically during the denoising process. Unlike prompt engineering, which is static, these methods steer the generation trajectory step-by-step. Techniques like Distribution Guidance (Parihar et al., 2024) and Fairgen (Kang et al., 2025) analyze the latent image state during inference; if the trajectory drifts toward a biased mode, the method applies a corrective gradient to steer it back toward a target distribution. Semantic editing frameworks like FairDiffusion (Friedrich et al., 2025) and cross-attention control (Hertz et al., 2022; Brack et al., 2023) allow for precise attribute swapping (e.g., changing gender) while preserving the scene's composition. Responsible Diffusion Framework (Azam and Akhtar, 2025) alters the conditioning in between the denoising process to achieve demographic diversity in the generated images.

Trade-off: While these methods offer granular control without retraining, they significantly increase inference latency due to the additional gradient calculations required at each denoising step (Meng et al., 2021).